Design and Implementation of a Broadband Low-Noise Inverter-Based Transimpedance Amplifier

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***Abstract*—High-speed optical and wireline communication systems demand transimpedance amplifiers (TIAs) that simultaneously achieve high gain, wide bandwidth, and low input-referred noise under strict power and area constraints. In nanometer CMOS technologies, parasitic capacitances associated with the photodiode and input devices severely limit bandwidth and noise performance. This paper presents a broadband inverter-based TIA architecture that exploits the high transconductance efficiency of complementary CMOS devices. Bandwidth enhancement is achieved using active feedback and inductive peaking techniques, while maintaining low noise and compact area. The proposed design demonstrates improved gain-bandwidth trade-offs and is suitable for multi-tens-of-Gb/s receiver front-end applications.**

***Index Terms*—Transimpedance amplifier, inverter-based amplifier, active feedback, inductive peaking, broadband CMOS, optical receivers**

I. Introduction

The rapid growth of high-speed data communication has driven the need for receiver front-end circuits capable of operating at data rates in the tens of gigabits per second. Optical interconnects, enabled by low-loss optical fibers, are widely adopted due to their superior bandwidth and immunity to electromagnetic interference. In such systems, the transimpedance amplifier (TIA) serves as the first electronic block following the photodiode and plays a critical role in determining overall receiver sensitivity, bandwidth, and noise performance.

Conventional TIA topologies such as common-gate and shunt-feedback amplifiers exhibit inherent trade-offs between input impedance, noise, and bandwidth. While common-gate TIAs offer wide bandwidth due to low input impedance, they typically suffer from higher noise. Shunt-feedback TIAs provide lower noise and higher gain but are limited in bandwidth by input capacitance. Inverter-based TIAs have recently gained attention due to their higher effective transconductance, low supply voltage operation, and improved gain efficiency. However, bandwidth extension remains a key challenge in deeply scaled CMOS technologies.

This work explores an inverter-based TIA architecture employing active feedback and inductive peaking techniques to overcome bandwidth limitations while preserving low noise and high gain.

II. Modulator Design and Implementation

The proposed TIA architecture is based on a CMOS inverter gain stage, which leverages both NMOS and PMOS devices to achieve higher transconductance compared to single-ended topologies. A shunt-feedback resistor is employed to set the transimpedance gain and stabilize the operating point, enabling self-biased operation at mid-supply for improved linearity.

To extend the bandwidth beyond the intrinsic pole imposed by the photodiode and input capacitances, an active feedback (AFB) technique is introduced. The active feedback path utilizes a scaled inverter stage, allowing controlled gain peaking at higher frequencies. By carefully sizing the inverter stages, multiple complex-conjugate poles are created, providing additional degrees of freedom to optimize bandwidth, gain flatness, and stability.

In addition to active feedback, inductive peaking techniques are explored to further enhance bandwidth. Series and shunt inductive elements compensate for capacitive loading at the input and output nodes, introducing zeros that extend the high-frequency response. The combination of inverter-based amplification, active feedback, and inductive peaking enables a compact, low-power TIA suitable for broadband operation.

III. Experimental Results and System Demonstration

The proposed TIA designs were implemented in a scaled CMOS technology and evaluated through post-layout simulations and measurements. Scattering parameters were used to characterize input matching, gain, and bandwidth performance. The photodiode was modeled using an equivalent capacitance to emulate realistic operating conditions.

Measurement results demonstrate a flat transimpedance gain over a multi-decade frequency range, with bandwidths exceeding several tens of gigahertz. Input and output return losses indicate adequate impedance matching, while noise analysis confirms that the dominant contribution arises from the feedback resistor at low frequencies. Compared to conventional shunt-feedback TIAs, the proposed architectures achieve superior gain-bandwidth trade-offs with reduced power consumption and silicon area.

IV. Conclusion

This paper presented a broadband inverter-based transimpedance amplifier architecture employing active feedback and inductive peaking techniques for high-speed receiver applications. By exploiting the high transconductance efficiency of CMOS inverters and introducing controlled gain peaking, the proposed design overcomes bandwidth limitations imposed by parasitic capacitances. Experimental results validate the effectiveness of the approach, demonstrating high transimpedance gain, wide bandwidth, and low noise in a compact implementation. The proposed TIA is well-suited for next-generation optical and wireline communication systems.